

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 1.5895486487212596, median 1.0350707308573794, std: 1.6661895330754368
Gyroscope error (imu0): mean 1.8438191352841302, median 0.9130273433276461, std: 2.2314023027500207
Accelerometer error (imu0): mean 1.1752541393446516, median 0.9627872091470302, std: 0.8865452687212337

Residuals

Reprojection error (cam0) [px]: mean 1.5895486487212596, median 1.0350707308573794, std: 1.6661895330754368
Gyroscope error (imu0) [rad/s]: mean 0.019368504715379903, median 0.00959094851881295, std: 0.023439894507908645
Accelerometer error (imu0) [m/s^2]: mean 0.12220759252860255, median 0.10011443738696427, std: 0.09218649765271983

Transformation (cam0):

T_ci: (imu0 to cam0):
[[0.99792923 0.06336637 0.01104295 -0.00006595]
[0.06350479 -0.99790109 -0.01267081 0.00008568]
[0.01021687 0.01334585 -0.99985874 -0.16103663]
[0. 0. 0. 1.]]

T_ic: (cam0 to imu0):
[[0.99792923 0.06350479 0.01021687 0.00170566]
[0.06336637 -0.99790109 0.01334585 0.00223885]
[0.01104295 -0.01267081 -0.99985874 -0.16101207]
[0. 0. 0. 1.]]

timeshift cam0 to imu0: [s] ($t_{imu} = t_{cam} + \text{shift}$)
-0.13680623812145945

Gravity vector in target coords: [m/s^2]
[-9.59609959 0.05564982 -2.01995018]

Calibration configuration

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cam0

Camera model: pinhole

Focal length: [458.22360596596207, 462.6623901963237]

Principal point: [316.55722409523264, 232.7935465992999]

Distortion model: radtan

Distortion coefficients: [0.07262855091689971, -0.17497410236495275, -0.004073105538382361, 0.000347733267084456]

Type: aprilgrid

Tags:

Rows: 6

Cols: 6

Size: 0.088 [m]

Spacing 0.026399999999999996 [m]

IMU configuration

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IMU0:

Model: calibrated

Update rate: 200

Accelerometer:

Noise density: 0.007352777114032854

Noise density (discrete): 0.10398397115771767

Random walk: 0.001032209763127813

Gyroscope:

Noise density: 0.0007427844067568089

Noise density (discrete): 0.010504557819547329

Random walk: 0.00012304514548225722

T_ib (imu0 to imu0)

[[1. 0. 0. 0.]

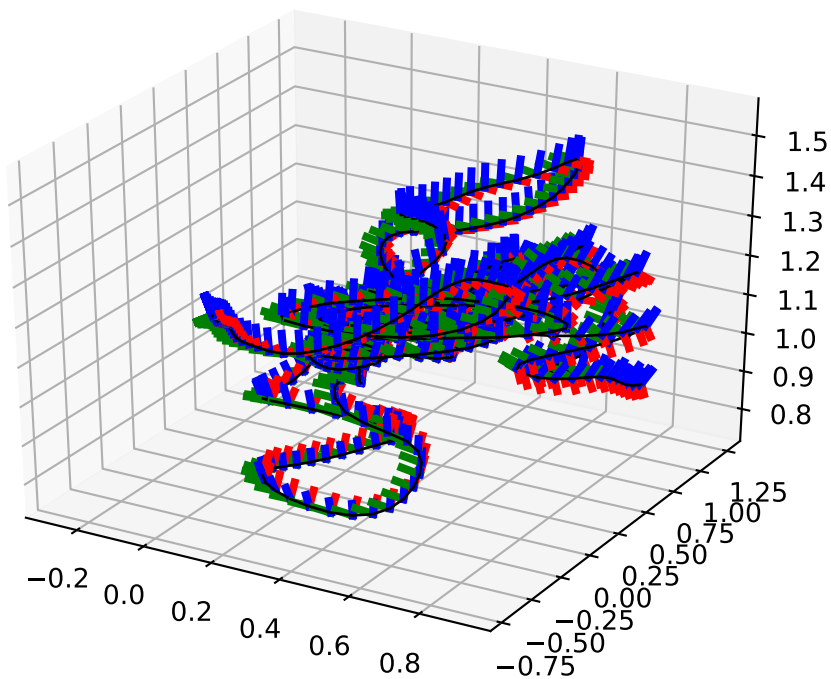
[0. 1. 0. 0.]

[0. 0. 1. 0.]

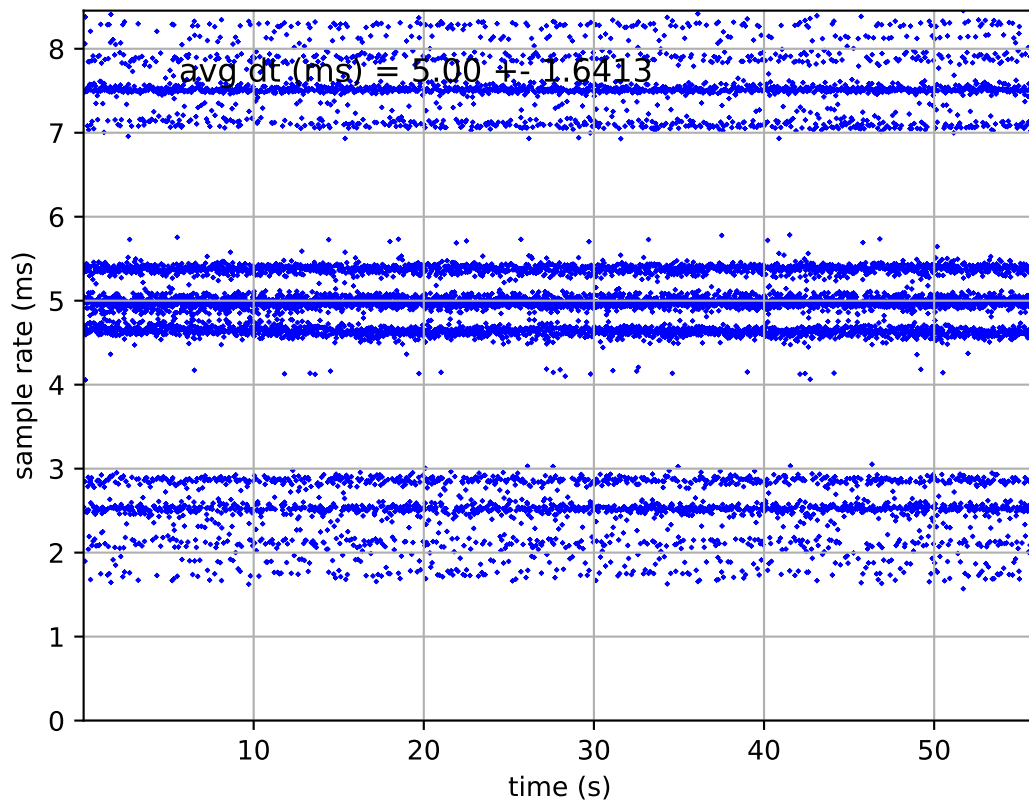
[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

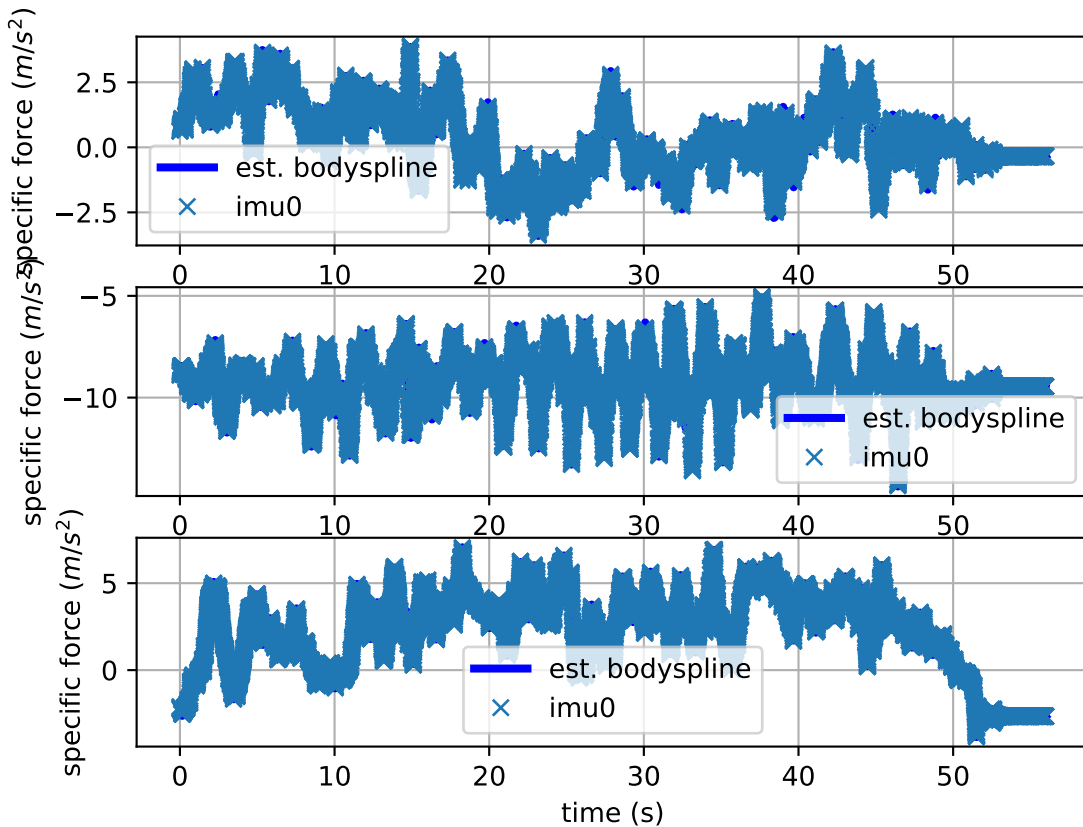
imu0: estimated poses



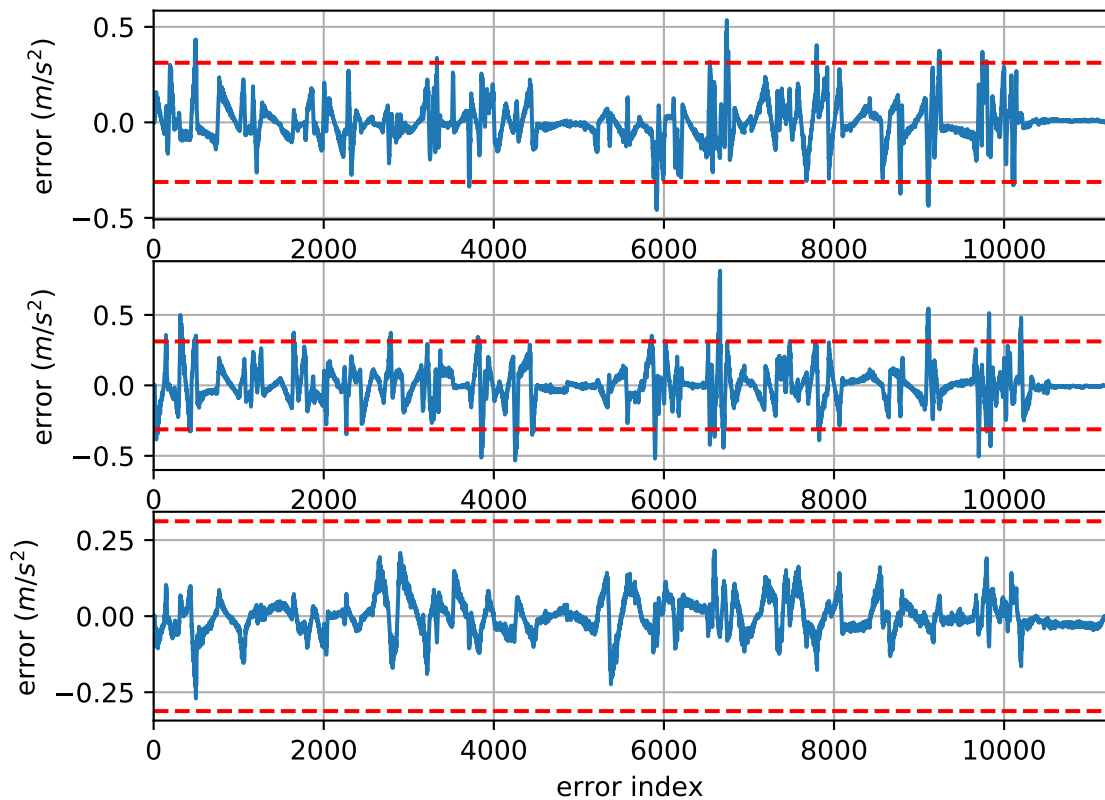
imu0: sample inertial rate



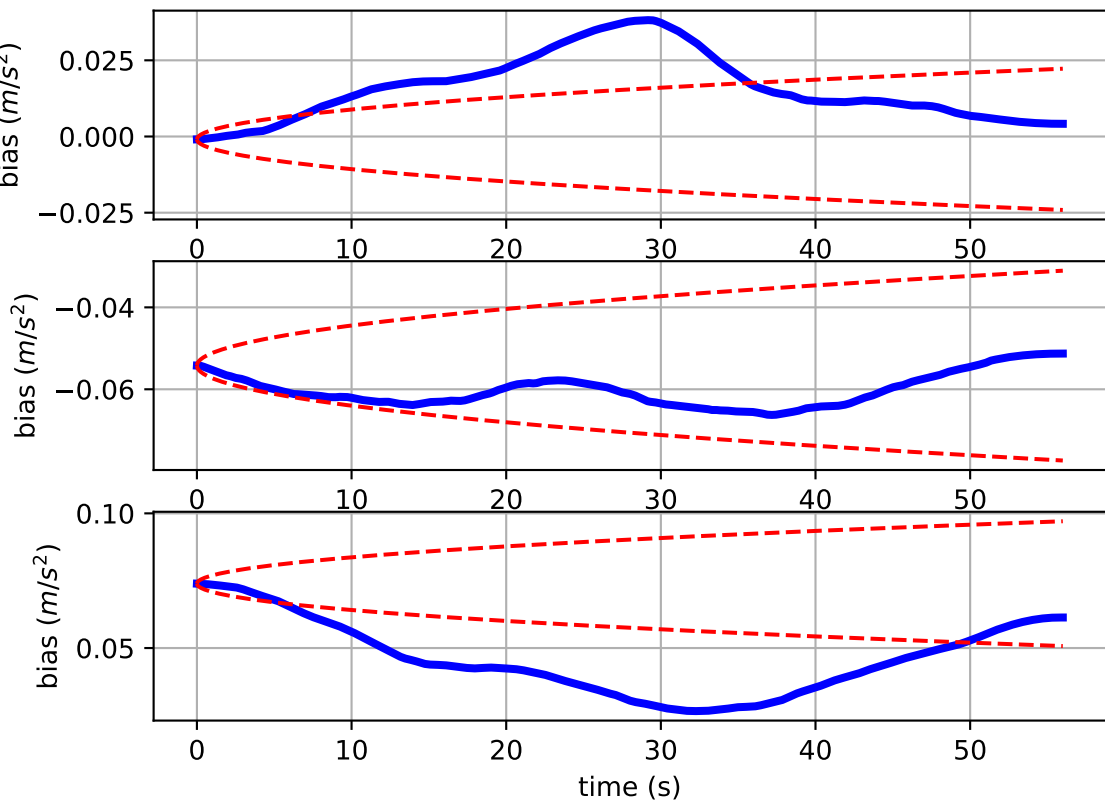
Comparison of predicted and measured specific force (imu0 frame)



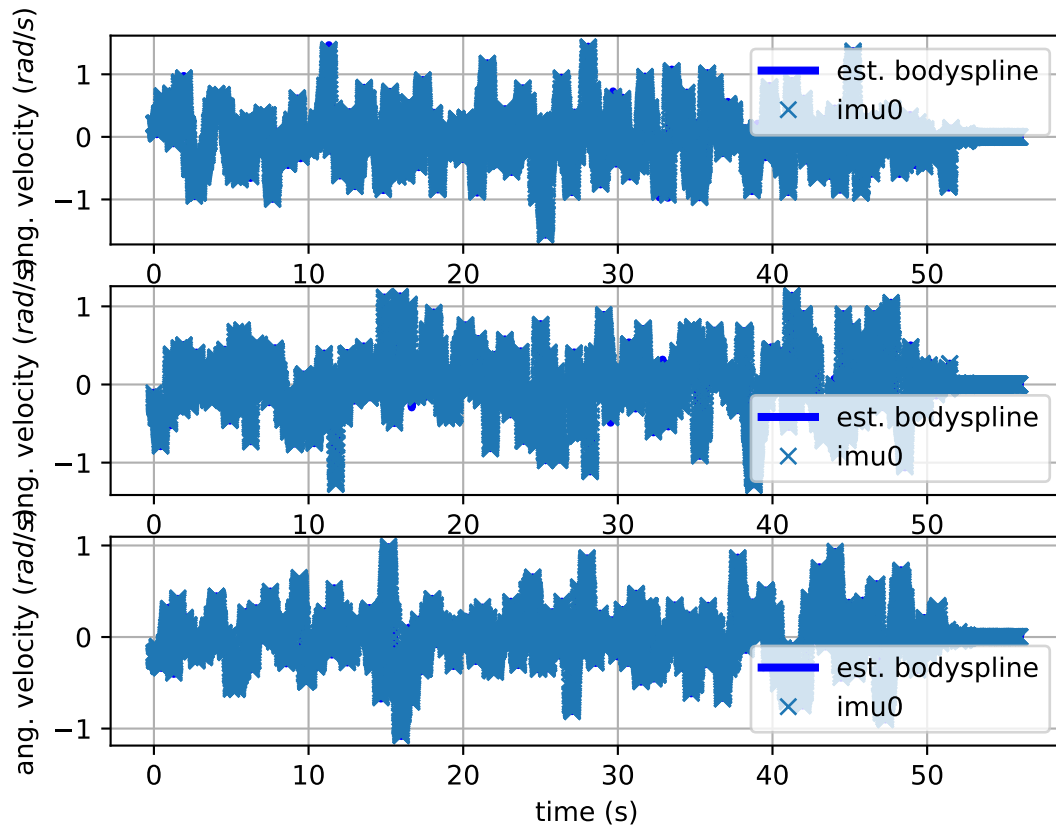
imu0: acceleration error



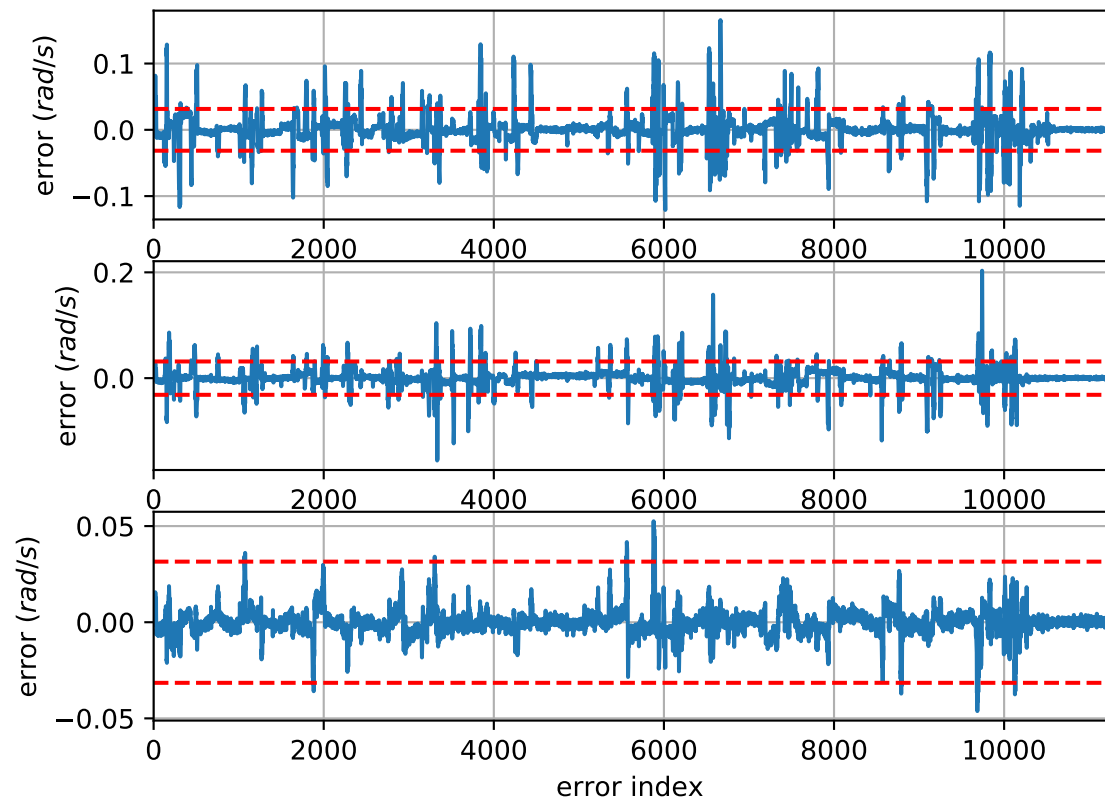
imu0: estimated accelerometer bias (imu frame)



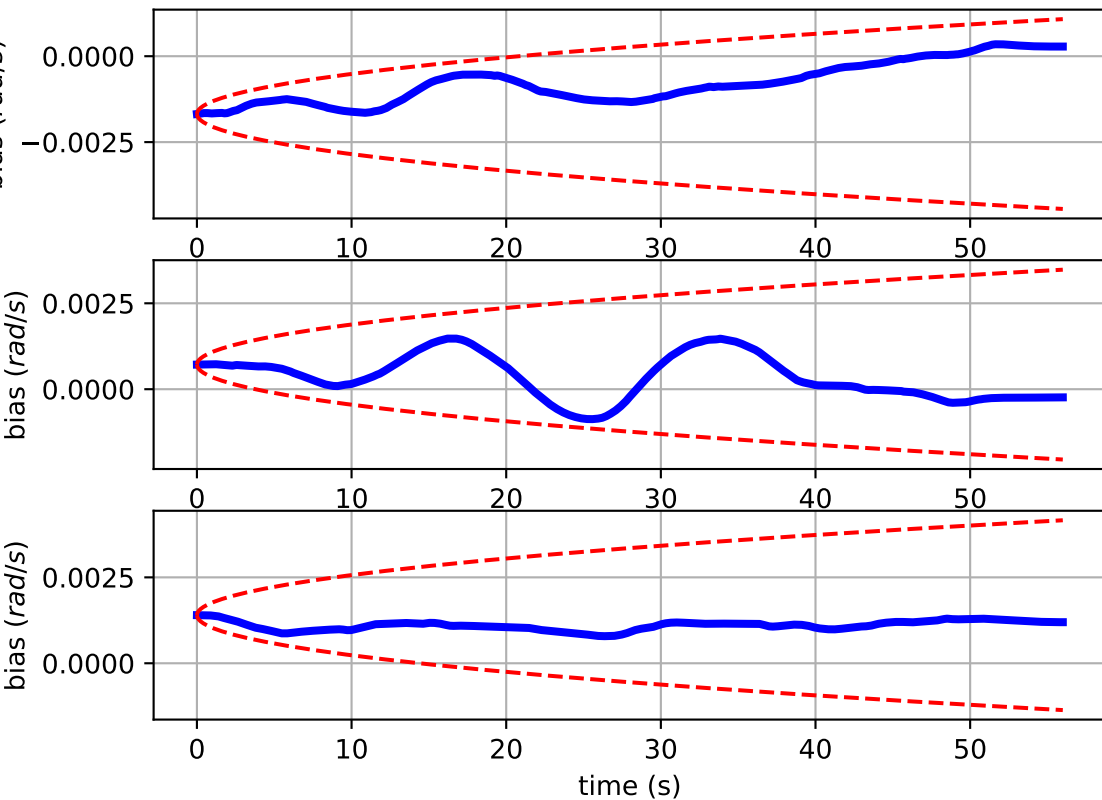
Comparison of predicted and measured angular velocities (body frame)



imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors

